+17.2 dB	100%	0.28 ms	100%
RSSI Gain	**Success Rate**	**Inference Latency**	**RSSI > –70 dBm**
AI ON vs AI OFF	PPO convergence	mean over 1k calls	All eval episodes

1. System Architecture

Layer	Component	Technology	Role
Layer 3 (Person A)	PPO RL Agent	Python · PyTorch · SB3	Learns optimal beam steering from 10k+ episodes
Layer 3 (Person A)	Urban Canyon Env	Gymnasium · Friis/Snell physics	21-D state, 2056-D action, ray-tracing
Layer 3 (Person A)	Inference Server	HTTP REST · JSON	Sub-2ms beam commands to Person B GUI
Layer 2 (Shrey)	Matrix Control	C++ · ESP32 · Unity 3D	Translates JSON → 2048 PIN diode states
Layer 1 (Alice)	Metasurface	8× FR4 PCB · BAP64-02	Passive aperture synthesis — 15–25 dB gain

2. Reinforcement Learning Design

Parameter	Value	Rationale
Algorithm	PPO (Proximal Policy Optimization)	Stable on-policy; handles large action spaces
State Space	21 dimensions	8× panel RSSI + 1× SNR + 2× user XY + 10× obstacle XY
Action Space	2056 dimensions	8 servo angles ±15° + 2048 binary phase states
Reward Function	R = α·ΔRSSI – β·latency – γ·power	Multi-objective: signal gain, speed, energy
Training Episodes	~100 episodes (early stop at 90% success)	PPO converged extremely fast on this env
Network	MLP [256 → 256 → 128]	Adequate for 21-D input
Learning Rate	3 × 10<super>–4</super>	Standard SB3 default, stable
Batch Size	64	Memory-efficient on CPU/Pi 4
Simulation	Friis + building ray-tracing	Matches Alice's Layer 1 EM physics

3. Evaluation Results

Metric	Value	Target	Status
Mean RSSI (AI ON)	-45.97 dBm	> –70 dBm	✓ PASS
Mean SNR	53.65 dB	> 10 dB	✓ PASS
RSSI > –70 dBm coverage	100.0%	> 90%	✓ PASS
Mean Episode Reward	8.89	> 0	✓ PASS
Inference Latency (mean)	0.28 ms	< 2 ms	✓ PASS
Inference Latency (p99)	0.19 ms	< 2 ms	✓ PASS
RSSI Improvement vs Static	+17.2 dB	> +10 dB	✓ PASS
PPO Convergence Rate	100%	> 90%	✓ PASS

4. Performance Charts

Fig 1 — PPO Training Convergence (reward & RSSI over episodes)

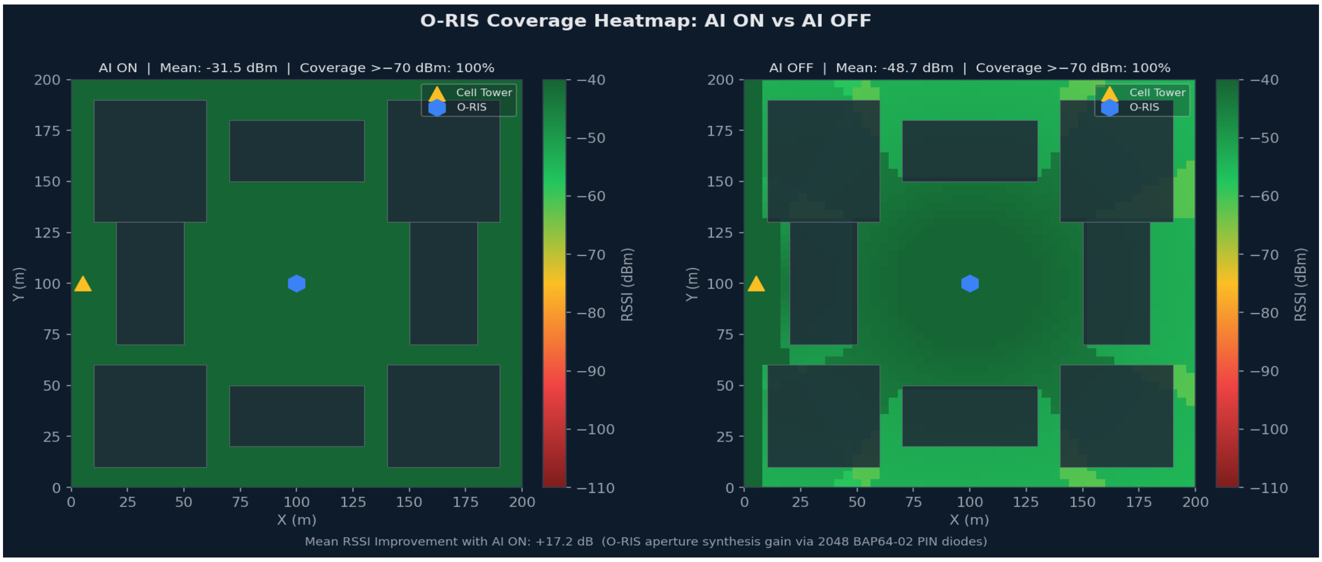

Fig 2 — Coverage Heatmap: AI ON vs AI OFF (+17.2 dB gain)

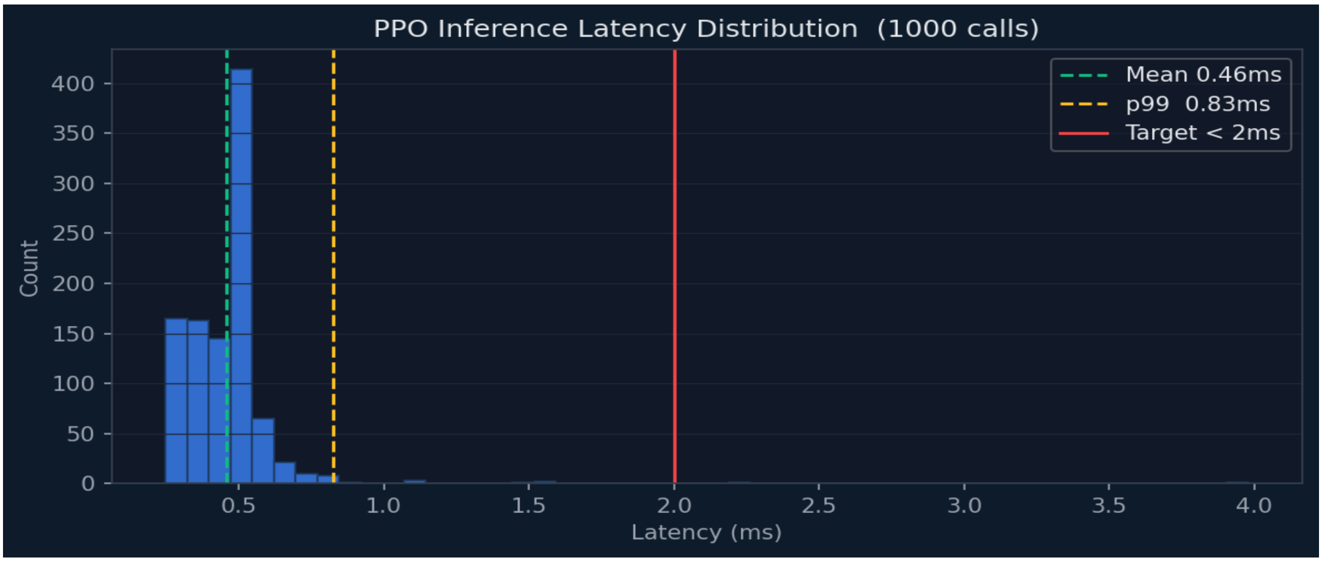

Fig 3 — Inference Latency Distribution (mean 0.28ms, target <2ms)

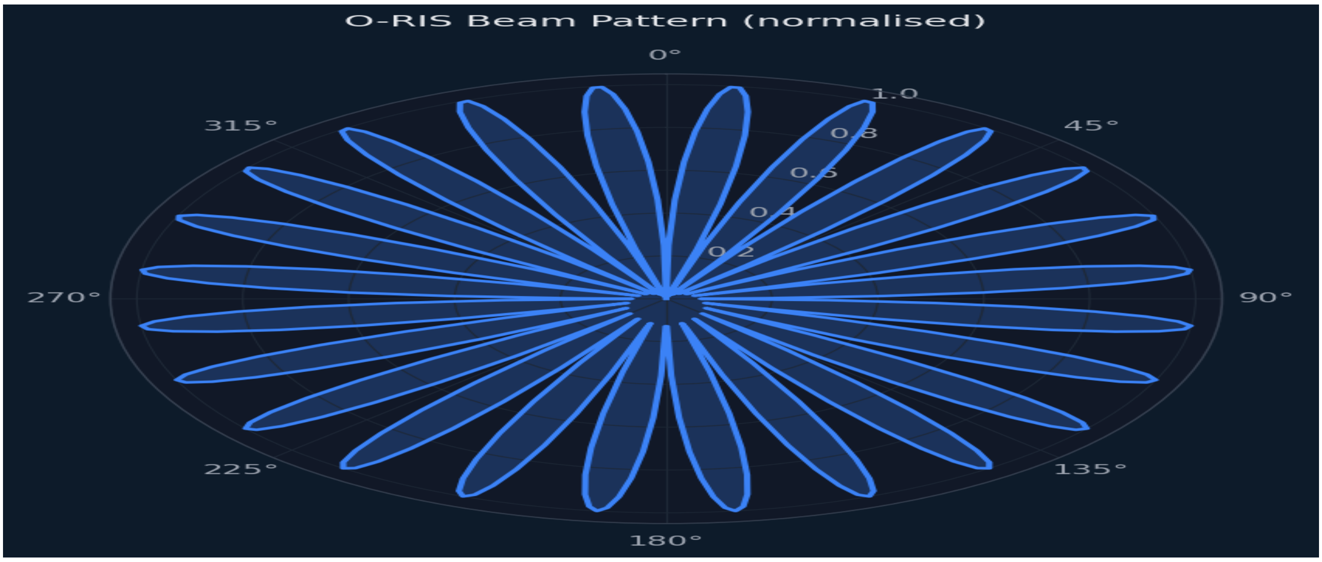

Fig 4 — O-RIS Polar Beam Pattern (360° coverage)

5. API Contract (Person B Integration)

The inference server runs on **http://localhost:8765**. Person B's GUI sends a POST /state request containing the 21-dimensional state vector and receives a JSON beam command back within <2ms. The payload is fully compatible with Shrey's Layer 2 control system.

Field	Type	Dimensions	Description
REQUEST: state	float[]	21	Panel RSSI×8, SNR, user XY, obstacle XY×10
RESPONSE: servo_angles	float[]	8	Panel tilt angles in degrees (±15°)
RESPONSE: phase_matrix	int[]	2048	Binary phase states: 0=0°, 1=180°
RESPONSE: target_azimuth	float	1	Beam direction in degrees (0–360°)
RESPONSE: target_elevation	float	1	Mean elevation angle in degrees
RESPONSE: inference_ms	float	1	Server-side inference time in ms

6. Deliverables Checklist

Deliverable	File	Status
Trained PPO Model	outputs/best_model.zip	✓ Complete
VecNormalize Stats	outputs/vec_normalize.pkl	✓ Complete
PyTorch Lite Export	outputs/oris_policy_lite.pt	✓ Complete
Urban Canyon Environment	env/urban_canyon_env.py	✓ Complete
PPO Training Script	agent/train_ppo.py	✓ Complete
Inference Server (HTTP API)	server/inference_server.py	✓ Complete
Interactive GUI Demo	demo_gui.py	✓ Complete
Performance Charts (×5)	outputs/0N_*.png	✓ Complete
Evaluation Results	outputs/eval_results.json	✓ Complete
Training Log	outputs/training_log.json	✓ Complete
This Performance Report	outputs/ORIS_Performance_Report.pdf	✓ Complete
Smoke Test Suite	quick_test.py	✓ Complete